Functional Forecasting of Turkish Electricity Load Curves

Thesis proposal v3

Object	Daily curve $Y_d(h),\ h = 0, \ldots, 23$
Main model	K=2 FPCA–VAR score model with holiday, load-shape, derivative features
Benchmark	Official day-ahead forecast published by TEİAŞ / EPIAS
Evaluation	Rolling origin, 2023–2025, 1,096 days, 26,304 hourly errors
Test	Diebold–Mariano, pooled and hour-by-hour

Zeynep Bozoklu PhD proposal

Forecasting Question

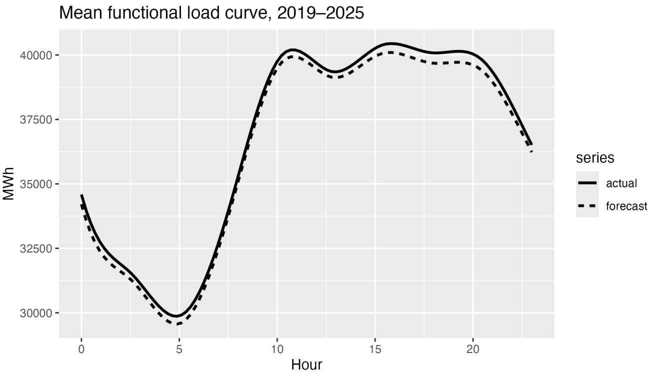
Operational question Can a functional statistical model forecast tomorrow's full 24-hour load profile more accurately than the official benchmark, using only information available before the target day?

Descriptive starting point The official forecast is systematically below realized load over the evaluation window. The proposed model reduces the average hourly bias by roughly four-fifths.

Year	Official bias MWh/hour	Main model bias MWh/hour
2023	344	176
2024	585	34
2025	472	98
Average	**467**	**103**

Positive bias: actual > forecast.

Forecasting Object: One Curve, Not 24 Unrelated Targets



$$Y_d = \{Y_d(h) : h = 0, \ldots, 23\}$$

- Hourly observations form a coherent daily trajectory.
- Level, peak contrast, and ramp speed are curve-level features.
- FDA reduces a 24-dimensional problem to a small number of scores.

Solid = actual mean curve; dashed = official mean forecast.

Research Design

Hourly load $Y_{d0}, \dots, Y_{d23}$ → B-spline curve → FPCA scores → Score forecast → Curve reconstruction

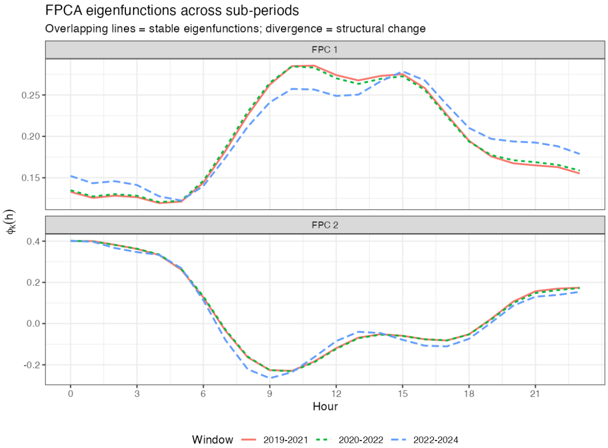
$$Y_d(h) \approx \mu(h) + \sum_{k=1}^{K} \xi_{dk}\phi_k(h)$$

$$\hat{Y}_d(h) = \hat{\mu}(h) + \sum_{k=1}^{K} \hat{\xi}_{dk}\hat{\phi}_k(h)$$

Design principles:

- low-dimensional representation;
- expanding-window estimation;
- official forecast as external benchmark;
- formal forecast-accuracy testing.

FPCA Compression and Stability



FPC	Variance	Cumulative
1	93.4%	93.4%
2	3.9%	97.3%
3	1.1%	98.4%
4	1.0%	99.4%

FPC 1	level
FPC 2	shape contrast
Stability	IP > 0.99
Robustness	$K = 4$ unchanged

Score Forecasting Model

$$\xi_d = a + A_1 \xi_{d-1} + A_7 \xi_{d-7} + B z_d + u_d$$

$$z_d \in \mathcal{F}_{d-1}$$

Estimated equation-by-equation by OLS. The target is the score vector, not an individual hour.

Block	Variables
Score dynamics	lag 1, lag 7
Calendar	weekend, annual sine/cosine
Holiday	public holidays, Eid, bridge days
Load shape	lagged mean, peak, range
Rolling history	7-day, 28-day averages
Derivative	ramp summaries, slope volatility

$\mathcal{F}_{d-1}$ denotes the information set before target day d; lagged curves are treated as known.

Rolling-Origin Evaluation and DM Test

For each target day d:

$$\{1, \dots, d-1\} \quad \Longrightarrow \quad \hat{Y}_d(0{:}23)$$

Metrics pooled over all target days and hours:

MAE, MAPE

Diebold–Mariano loss differential:

$$\Delta_t = |e_t^{\text{model}}| - |e_t^{\text{official}}|$$

$$H_1 : \mathbb{E}[\Delta_t] < 0$$

HAC variance and Harvey–Leybourne–Newbold correction.

Same test is also run separately for each hour h:

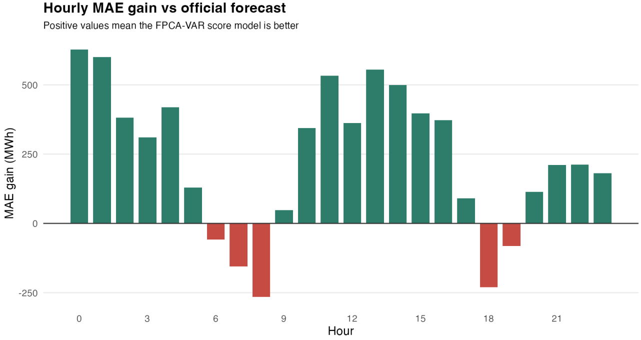
$$\Delta_{d,h} = |e_{d,h}^{m}| - |e_{d,h}^{o}|.$$

Main Empirical Result

Model	MAE	MAPE	Gain vs official	DM
Seasonal naive Y_{d-7}	1912	5.13%	-57.6%	–
Official TEİAŞ forecast	1214	3.15%	ref.	–
FPCA–VAR scores only	1443	3.76%	-18.9%	worse
+ calendar / ramp	1238	3.26%	-2.1%	n.s.
+ holiday / load-shape	1001	2.63%	$+17.5\%$	***
+ **derivative features**	**980**	**2.56%**	$+19.2\%$	***
Residual second step	653	1.70%	$+46.2\%$	***

Evaluation: 2023–2025. Main model pooled DM statistic $= -27.4$, $p \approx 1.3 \times 10^{-163}$. MAE in MWh/hour.

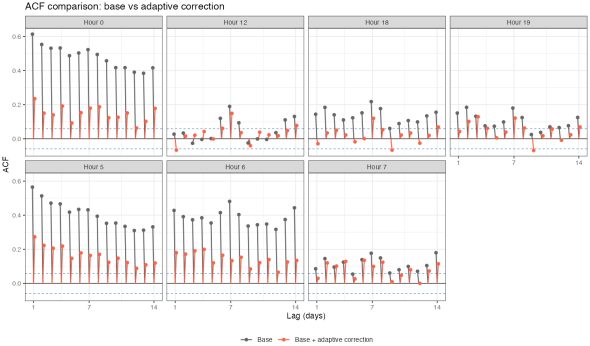
Hourly Diagnostics for the Main Model



Diagnostic	Count
Lower MAE	19 / 24
DM-significant	18 / 24
Weak window	6–9
Strong gains	0–5, 10–17

Positive bars indicate lower MAE than the official benchmark. The table distinguishes raw wins from statistically significant wins.

Residual Second Step



Base-model residuals are serially structured:

$$r_d(h) = Y_d(h) - \hat{Y}_d^{\text{base}}(h)$$

Correction:

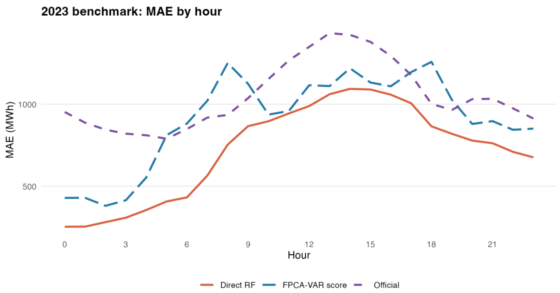
$$\hat{Y}_d^{\text{res}}(h) = \hat{Y}_d^{\text{base}}(h) + \lambda_d \hat{r}_d(h)$$

λ_d chosen by rolling-origin CV

using pre-d data only.

Result: MAE 653, MAPE 1.70%, gain $+46.2\%$, lower MAE at 24/24 hours.

Random Forest Benchmark



Benchmark sample: 2023, 8,760 hourly forecasts.

Model	MAE	Gain	DM
Official	1052	ref.	–
FPCA score	910	$+13.5\%$	***
Direct RF	718	$+31.8\%$	***

Direct RF predicts each hour directly and bypasses FPCA. It has lower MAE at 24/24 hours on this sample; pairwise DM vs the FPCA–VAR score model: $p \approx 1.2 \times 10^{-85}$.

FPCA score = main K=2 model without residual correction, restricted to the same 2023 benchmark sample.

What Each Comparison Answers

Comparison	Statistical question	Interpretation
Main FPCA–VAR vs official	Does a transparent functional score model improve the operational benchmark?	Yes: $+19.2\%$ MAE, pooled DM $p \approx 10^{-163}$.
Residual step vs official	Are base-model errors predictably autocorrelated?	Yes: residual correction wins at all hours.
Direct RF vs official	How strong is a direct nonlinear hourly benchmark on the 2023 sample?	Very strong: $+31.8\%$ MAE, DM significant.
Direct RF vs FPCA–VAR	Does the direct nonlinear benchmark beat the interpretable score model on the same 2023 sample?	Yes: pairwise DM $p \approx 1.2 \times 10^{-85}$.

Thesis position The FPCA–VAR model is the interpretable functional baseline; Direct RF is a serious benchmark that documents remaining nonlinear predictive structure.

Contributions and Status

Contribution	Evidence in current draft
Functional representation	B-spline curves; $K = 2$ FPCA retains 97.3%; stable eigenfunctions.
Forecasting model	Expanding-window FPCA–VAR score model with predetermined feature blocks.
Formal benchmark testing	Pooled and hour-specific Diebold–Mariano tests against the official forecast.
Residual diagnostics	Autocorrelation motivates a no-lookahead residual second step.
Nonlinear benchmark	Direct RF reported separately, with MAE and DM comparisons.

Done empirical pipeline, main results, RF benchmark, DM diagnostics

Open systematic literature review, final write-up, scope decision on intervals

References

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